Revision Game Cards — 1.3 Exchanging Data

Print and cut along the borders. ✂️ Loop cards: each player reads the bottom "Who has…?"; whoever holds the matching "I have…" reads it out, then their own clue — the chain visits every card and loops back to the start.

Loop cards (dominoes) — cut out & deal

1 / 16

I HAVE

RLE

WHO HAS…

The lossless method that stores repeated patterns once and references them?

2 / 16

I HAVE

Dictionary compression

WHO HAS…

Compression where data is permanently removed and cannot be recovered?

3 / 16

I HAVE

Lossy compression

WHO HAS…

The one-way function that produces a fixed-length digest?

4 / 16

I HAVE

Hashing

WHO HAS…

The encryption type that uses the same key to encrypt and decrypt?

5 / 16

I HAVE

Symmetric encryption

WHO HAS…

The encryption type that uses a public and private key pair?

6 / 16

I HAVE

Asymmetric encryption

WHO HAS…

The attribute(s) that uniquely identify each record in a table?

I HAVE

Primary key

WHO HAS...

The primary key of another table used to link tables together?

I HAVE

Foreign key

WHO HAS...

The process of organising data to reduce redundancy and anomalies?

I HAVE

Normalisation

WHO HAS...

The rule requiring no partial dependencies on the whole key?

I HAVE

2NF

WHO HAS...

The properties guaranteeing reliable transactions: Atomicity, Consistency, Isolation, Durability?

I HAVE

ACID

WHO HAS...

The network where data is split into packets routed independently and reassembled?

I HAVE

Packet switching

WHO HAS...

The system that translates domain names into IP addresses?

I HAVE

DNS

WHO HAS...

The device that connects different networks together using IP addresses?

I HAVE

Router

WHO HAS...

The device that connects devices within a single LAN using MAC addresses?

I HAVE

Switch

WHO HAS…

The algorithm that ranks web pages by importance based on incoming links?

I HAVE

PageRank

WHO HAS…

The lossless method that replaces runs of repeated values with a value and a count?

Taboo cards — describe the term without the banned words

Hashing

Don't say:
- one-way
- password
- digest
- reversible

RLE

Don't say:
- run
- repeated
- count
- lossless

Foreign key

Don't say:
- primary
- link
- table
- reference

Normalisation

Don't say:
- redundancy
- anomaly
- table
- 1NF

Packet switching

Don't say:
- split
- route
- reassemble
- internet

DNS

Don't say:
- domain
- IP
- address
- translate

Symmetric encryption

Don't say:
- same
- key
- decrypt
- distribution

PageRank

Don't say:
- link
- importance
- vote
- ranking

Bingo — word bank & ready-to-cut cards

Word bank (students fill a blank 3×3 grid, or use the pre-filled cards below): Lossless, Lossy, RLE, Dictionary compression, Symmetric, Asymmetric, Hashing, Primary key, Foreign key, Composite key, Normalisation, ACID, Record locking, LAN, WAN, Packet switching, DNS, Router, Switch, PageRank

BINGO 1

LAN	Dictionary compression	Record locking
ACID	RLE	Primary key
Packet switching	Foreign key	Hashing

BINGO 2

Lossy	Hashing	RLE
Record locking	Asymmetric	Switch
ACID	Packet switching	Composite key

BINGO 3

DNS	LAN	Primary key
Symmetric	PageRank	Packet switching
Router	RLE	Normalisation

BINGO 4

Packet switching	Router	ACID
Primary key	WAN	DNS
Foreign key	PageRank	Hashing

BINGO 5

LAN	Dictionary compression	Router
DNS	Hashing	Record locking
PageRank	WAN	Switch

BINGO 6

Packet switching	Record locking	Router
LAN	Composite key	Asymmetric
Foreign key	Normalisation	RLE

Bingo — caller's list (teacher: read the clue, students cross off the term)

#	Clue to read aloud	Answer
1	Compression where the original can be recovered exactly.	Lossless
2	Compression that permanently removes data, used for JPEG and MP3.	Lossy
3	Replaces runs of repeated values with a value and a count.	RLE
4	Stores repeated patterns once and references them; the index must be sent too.	Dictionary compression
5	Encryption using the same key to encrypt and decrypt.	Symmetric
6	Encryption using a public and private key pair.	Asymmetric
7	One-way function producing a fixed-length value; used to store passwords.	Hashing
8	Attribute(s) that uniquely identify each record.	Primary key
9	The primary key of another table, used to link tables.	Foreign key
10	A primary key made from two or more attributes combined.	Composite key
11	Organising data to reduce redundancy and remove anomalies.	Normalisation
12	The four properties of a reliable transaction.	ACID
13	Locking a record during a transaction to prevent the lost update problem.	Record locking
14	A network covering a small area on a single site, hardware owned by one organisation.	LAN
15	A network covering a large geographic area, using third-party infrastructure.	WAN
16	Data split into independently routed units, reassembled at the destination.	Packet switching
17	Translates domain names into IP addresses.	DNS
18	Connects different networks together using IP addresses.	Router
19	Connects devices within a LAN using MAC addresses.	Switch
20	Ranks web pages by importance using incoming links.	PageRank

Quick-fire — clue & answer (self-test / quiz-quiz-trade)

#	Clue	Answer
1	The lossless compression best suited to images with long runs of identical pixels.	RLE (Run Length Encoding)
2	The encryption weakness that asymmetric encryption solves by sharing a key openly.	Key distribution problem
3	The property of a hash function meaning the same input always gives the same output.	Deterministic
4	The normal form that removes transitive dependencies.	3NF (Third Normal Form)
5	The table used to resolve a many-to-many relationship.	Linking / junction table
6	The ACID property meaning a transaction is all-or-nothing.	Atomicity
7	The rule that a foreign key must reference a valid existing primary key.	Referential integrity
8	The SQL keyword that filters which rows are returned.	WHERE
9	The SQL clause that combines columns from related tables on a matching key.	JOIN (... ON)
10	The TCP/IP layer that adds source and destination IP addresses.	Network / Internet layer
11	The protocol used only for sending email.	SMTP
12	JavaScript form validation runs in this location, which is why it is not secure on its own.	Client-side (the browser)